

QITAI, China

WMO: 513790

Lat: 43.9953N

Lon: 89.6044E

Elev: 2608

StdP: 13.36

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-17.3	-28.4	1.2	-22.0	-22.5	1.7	-16.9	14.1	3.2	12.2	-0.9	5.9	160	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.6	93.9	64.1	91.2	63.3	88.5	62.4	67.1	86.2	65.6	84.6	64.4	83.2	4.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.3	89.1	71.9	59.6	84.0	70.9	58.0	79.1	70.9	33.1	85.5	31.9	84.7	30.9	83.0	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.4	13.5	11.4		-26.1	100.2	5.6	3.6	-30.2	102.7	-33.5	104.8	-36.6	106.9	-40.7	109.5	
				-26.4	69.9	5.6	2.8	-30.4	72.0	-33.6	73.6	-36.8	75.2	-40.8	77.2	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	2.6	9.2	30.9	51.4	62.3	70.8	73.7	71.6	61.6	46.8	29.0	9.3	
	DBStd	26.53	9.19	8.94	13.30	9.04	7.19	5.42	4.82	5.54	7.33	8.24	10.54	9.40	
	HDD50	5368	1468	1143	600	91	6	0	0	0	8	161	630	1261	
	HDD65	8646	1933	1563	1059	412	136	15	2	11	145	564	1080	1726	
	CDD50	2969	0	0	6	133	386	623	734	669	356	62	0	0	
	CDD65	774	0	0	0	4	51	188	271	215	44	1	0	0	
	CDH74	9931	0	0	0	133	833	2235	3188	2772	747	23	0	0	
	CDH80	4026	0	0	0	24	242	879	1434	1218	226	3	0	0	
Wind	WSAvg	5.7	4.8	4.9	5.6	7.1	6.7	6.4	5.9	5.8	5.7	5.3	5.1	4.7	
Precipitation	PrecAvg	6.9	0.2	0.2	0.4	0.7	0.7	1.0	1.0	0.6	0.8	0.6	0.4	0.3	
	PrecMax	12.8	0.6	0.7	0.9	1.9	1.9	2.4	3.1	2.0	2.2	1.4	1.3	0.7	
	PrecMin	3.5	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.1	0.0	0.0	0.1	0.0	
	PrecStd	1.9	0.2	0.2	0.2	0.4	0.5	0.6	0.8	0.5	0.6	0.3	0.2	0.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.7	35.3	68.1	82.9	89.7	95.2	99.2	97.3	91.2	76.9	60.6	34.8	
		MCWB	26.0	30.3	48.0	54.5	59.6	63.7	66.7	63.9	60.1	52.1	45.0	30.0	
	2%	DB	24.6	30.5	61.8	77.5	85.8	91.3	94.4	93.4	86.0	71.4	53.1	30.0	
		MCWB	22.3	26.8	46.8	51.7	58.2	63.4	65.4	63.3	58.6	50.5	41.4	27.5	
	5%	DB	20.4	27.1	56.3	73.3	82.7	88.5	91.3	90.4	82.0	67.0	48.0	26.4	
		MCWB	18.6	24.2	44.6	50.3	57.2	62.1	64.3	62.9	57.0	49.0	39.1	24.2	
	10%	DB	16.6	23.8	50.7	69.1	78.8	85.5	88.2	87.2	78.2	62.6	43.2	23.2	
		MCWB	15.0	21.5	41.9	48.7	55.4	61.5	63.6	61.9	55.7	47.5	36.6	21.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.9	31.2	50.2	56.1	62.7	67.7	70.6	67.7	62.8	53.8	45.7	30.9
			MCDB	28.9	34.4	64.3	78.6	84.0	87.9	91.4	85.2	82.5	74.0	58.9	33.6
2%		WB	22.4	27.2	47.4	53.6	60.0	65.7	67.9	65.8	60.2	51.6	42.0	27.7	
		MCDB	24.5	30.0	60.0	71.6	81.0	84.7	87.6	86.1	81.6	67.9	51.6	29.9	
5%		WB	18.6	24.4	44.9	51.7	58.2	64.3	66.4	64.5	58.4	49.7	39.5	24.5	
		MCDB	20.5	27.1	55.5	68.8	77.9	82.8	85.0	85.0	78.8	64.7	47.3	26.3	
10%		WB	15.1	21.5	42.2	50.0	56.6	62.9	65.1	63.2	56.6	47.9	37.0	21.6	
		MCDB	16.5	23.7	50.7	66.8	75.2	80.9	83.2	82.9	75.4	61.7	43.0	23.3	
Mean Daily Temperature Range	5% DB	MDBR	21.8	23.2	21.1	23.9	23.3	23.0	23.6	25.5	26.9	24.0	18.8	19.2	
		MCDBR	22.5	24.2	27.3	31.6	30.1	30.0	30.7	31.6	33.2	30.5	25.1	19.4	
		MCWBR	20.1	20.6	14.1	12.7	10.7	10.8	10.2	10.8	13.0	14.1	15.3	16.7	
	5% WB	MCDBR	21.8	22.9	25.1	25.7	25.5	23.6	25.2	27.0	28.6	27.8	23.7	18.0	
		MCWBR	19.5	19.5	13.4	10.9	9.8	9.3	9.4	9.8	11.6	13.2	14.7	15.6	
Clear-Sky Solar Irradiance	taub	0.332	0.367	0.412	0.459	0.454	0.471	0.452	0.428	0.401	0.385	0.353	0.327		
	taud	2.120	2.074	1.992	1.924	1.986	2.012	2.105	2.153	2.150	2.109	2.128	2.138		
	Ebn at Noon	239	252	256	255	260	256	259	261	258	242	228	227		
	Edh at Noon	31	40	49	57	56	54	49	45	42	38	31	28		
All-Sky Solar Radiation	RadAvg	625	939	1374	1802	2109	2168	2075	1915	1634	1148	714	518		
	RadStd	33	42	76	111	103	121	97	95	71	57	56	50		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)		+0.90	N/A	N/A	N/A	N/A	-1.08	N/A	N/A	+88

Nomenclature: See separate page